

G H Raisoni College of Engineering & Management, Pune (An Autonomous Institute affiliated to SP Pune University)
S.Y. B. Tech. /Semester-III /Div-A/2023-24
Department of Data Science
UDSL202: Database Management Systems

Question Bank

Unit I	BL	CO	PO
1 Explain the architecture of DBMS.	1	CO1	PO1
2 Draw and explain the detailed system architecture of DBMS.	2	CO1	PO1
3 Describe in detail about two-tier and three-tier client-server architecture of DBMS	1	CO1	PO1
4 Explain DBMS applications.	1	CO1	PO1
5 What is Data Base Administrator? Discuss the functions of DBA.	1	CO1	PO1
6 What is a data model? Explain different categories of data models	1	CO1	PO1
7 What is Entity set? and also define Relationship set. List and explain the symbols used to draw ER Diagram.	1	CO1	PO1
8 Define generalization and aggregation. Demonstrate generalization and aggregation using E-R diagram.	2	CO1	PO1
9 Differentiate specialization and generalization.	2	CO1	PO1
10 Discuss in detail about the concepts of E-R model with suitable examples.	2	CO1	PO1
11 Write about different types of attributes in ER model. Show the notation of each	1	CO1	PO1
12 Explain about various constraints used in ER-model.	1	CO1	PO1
13 How to represent a weak entity set in ER diagram? Quote suitable example.	1	CO1	PO1
14 Construct ER diagram for a car insurance company that has a set of customers each of Whom owns one or more cars. Each car has associated with it zero to any no. of recorded accidents.	3	CO1	PO1
Unit 2			
1 With the aid of relevant examples illustrate different DDL statements supported by SQL.	2	CO2	PO2
2 List and explain any 2 DML commands with syntax and example?	2	CO2	PO2
3 Write short notes on i) DDL ii) DML iii) Database Schema.	2	CO2	PO2
4 What is DML? Explain DML operations with examples.	1	CO2	PO2
5 Explain aggregate functions of SQL.	2	CO2	PO2
6 Explain any 5 string functions of SQL	2	CO2	PO2
7 Explain constraints in SQL.	2	CO2	PO2

8 Explain with examples in SQL 1. Create table 2. Insert Command 3. Update 4 Select 5. Alter	2	CO2	PO2
9 With the aid of relevant examples illustrate different DDL statements supported by SQL.	2	CO2	PO2
10 Illustrate different types of joins in SQL	2	CO2	PO2
11 Differentiate between where clause and group by clause.	2	CO2	PO2
12 Explain with example Use of a) group by, b) having, c) order by, d) join and its types, e) Exist, f) view	2	CO2	PO2
13 Discuss the following clauses with examples (i) HAVING (ii) GROUP BY (iii) Relational set operations	2	CO2	PO2
14 What is a join? Discuss different types of joins.	2	CO2	PO2
15 What is NULL? What is its importance? How are these values handled in relational model?	2	CO2	PO2
16 Write SQL Queries for following set of tables: EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, MnagerStartDate). i) Create tables ii) Display all employees in Department named 'Marketing'. iii) Display the name of highest salary paid 'female' employee	3	CO2	PO3
17 Create Table: Teacher_ info{Teacher_ id, Teacher_ Name, Dept_ Name, Sal, status} and Consider below values in respective fields of 'Teacher_ info' collection (Pic001, Ravi, IT, 30000, A) (Pic002, Aarti, IT, 40000, A) (Pic003, Narendra , COMP, 20000, B) (Pic004, Swati, ENTC, 15000, C) (Pic005, Sampada, IT, 35000, A) (Pic006, Mihir, IT, 25000, B) (Pic007, Malati, ENTC, 36000, A) and write SQL queries for following records. i) Display all contents of 'Teacher_ info'. ii) Display information of teacher whose salary is 35000. iii) Display information of teacher whose status is not equal to C. iv) Display information of teacher whose status is either A or Salary is 15000. v) Find teacher name whose name ends with letter 'i'. vi) Find Average salary of all teachers	3	CO2	PO3
18 EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, MnagerStartDate). i) Create tables	3	CO2	PO3

ii)	Display all employees in Department named 'Marketing'.			
iii)	Display the name of highest salary paid 'female' employee			